

Zygmunt Marchwian

Control Systems/**Integration/Software Engineer**

Los Angeles, CA
zygmunt.march@gmail.com
(310) 699-8860

Objective:

Control Systems/PLC/Integration/Software Engineer - only local opportunities to **CA 90034** or remote/hybrid.

Education:

AGH University of Science and Technology, Krakow, Poland:

- Masters of Automation and **Computer Science** and Electronics - 1981
- Completed Doctor's Degree Studies in Thermal **Engineering** - 1985

Capabilities:

Control systems design, **integration**, and upgrades:

- Proficient in **PLC engineering**, **programming**, **configuration**, **integration**, and **troubleshooting** platforms:
Beckhoff, Delta, IDEC, Keyence, Mitsubishi, Modicon - Schneider, Omron, OPTO 22, Panasonic, Rockwell, Siemens, Unitronics, Wago, and any IEC 61131-3 compatible
- Modular, object-oriented design based on customer specification – UML, P&IDs
- Hardware and **software** evaluation, testing, and support in selecting the optimal **configuration** based on control goals and budget
- OPC, MQTT, XML, **Modbus**, and other **communication** technologies
- Vision systems Cognex (VisionPro, InSight, Checker), QVI Measure X, QC5000, RFID, **Yamaha and Fanuc Robots**
- HMI design and **development**, **SCADA**, Condition Monitoring, Data Acquisition
- Motion Controls: Servo Drives, Steppers, VFDs, actuators
- QA/QC – Quality assurance/Quality control
- Documenting projects with AutoCAD **Electrical** and MS Office Suite (Visio – Flow charts)
- Commissioning, startup, and technical support

Software design using advanced Structured **Analysis**:

- Object, Component Oriented **Programming**
- **Programming skills** (VB, C, C#, ADO, SQL, XML, WPF and Visual Studio.NET, Python)
- Numerical modeling, **computer** simulations of systems and processes (LABVIEW, MATLAB), systems optimization and real-time systems **programming** and embedded systems
- General networking, protocols, **troubleshooting** – Wireshark **software**.

Have a gift for finding innovative solutions and for being able to integrate all aspects of bringing a concept to reality

Good **problem** solving, judgment, and decision-making **skills**

Management of internal personnel and outside contractors developing a **project**

Work History:

Evolution Construction, Sylmar, CA

June 2023 – March 2024 contract

- Project:** Automated servo system to transfer materials from multiple material storage locations to material processing stages. Materials database with selectable motion recipes.
- Title:** System Integration Engineer
- Function:** Design of the control system for the material handling gantry.
- PLC Delta AS300, HMI DOP-B, Servo ASDA-A2, CANopen communication
 - Delta DOPSoft 2.00.07, ISPSOFT 3.17 (Structured Text), ASDA_SOFT_V5.5.0.0
 - Visual Studio 2022 C#, Modbus TCP

Jabil Inc., Anaheim, CA

February 2023 – May 2023 contract

- Project:** The purpose of the concept and feasibility (C&F) Cell Build tool is to assemble multi-layer “dual unit cell” assemblies within a lab-scale system. This system shall act as a concept and feasibility system to path find acceptable tooling and methods for automated cell assembly.
- Title:** Software Design Engineer
- Function:** Design of the control software solution.
- Beckhoff TwinCAT 3, EtherCAT, and Beijer IO modules, VisualStudio 2017, C#, WPF, MVVM, Postgres Database, custom SCADA app
 - Denso VS-068A4/RC8A robot, 600002-0686 DENSO RC8A CONTROLLER, Cognex Vision Pro, Yaskawa SIGMA-7 EtherCAT servo,
 - NEOUSYS NUVO-7006E-I70CG9 system embedded PC

Energy Vault Inc., Westlake Village, CA

October 2021 – March 2022

- Project:** Energy Vault develops sustainable energy storage solutions designed to transform the world’s approach to utility-scale energy storage for grid resiliency.
- Title:** Senior Software Engineer
- Function:** Design of the safe motion system transferring heavy loads between storage nodes.
- PLC Siemens S7-1200, G120 CU250S-2, Profinet, Profisafe, TIA Portal v.17, VPN
 - VisualStudio2019, C#, SQL server

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Repligen, Rancho Dominguez, CA

August 2019 – May 2021 contract

August 2018 – January 2019 contract

- Project:** Spectrum® hollow fiber filters used in microfiltration and ultrafiltration applications. Automation of hollow fiber production with improved quality control.
- Title:** Senior Control Systems Engineer
- Function:** Based on the processed collected data designing and implementing a new program running on existing hardware to improve product quality and minimize downtime. Data acquisition, visual production monitoring, automated tests for quality control, 24/7 production support.
- CompactLogix, Panel View 6 Plus, C-more, Kinetix, PowerFlex, Oriental Steppers, Cognex, VPN
 - Rockwell Studio5000, FactoryTalk View ME, VisualStudio2019, C#, SQL server, QC5000

Cynvenio Biosystems, Inc., Westlake Village, CA

March 2019 – June 2019 contract

April 2016 – May 2018

- Project:** Liquid Biopsy Rare Cell Isolation Platform
- Title:** Principal Automation/Software Engineer
- Function:** Siemens PLC architecture update, PLC programming – SCL language, design the system of diagnostics and recipes – WinCC, continuous improvements for process and equipment.
- Siemens PLC S7-300, 1500, HMI TP700 Comfort, TIA PORTAL V13, 14, WinCC, SCL, PROFINET
 - PCON – IAI servo, steppers, RS485
 - Image data processing, Machine Learning, clusters, Visual Studio.NET 2017 - C#, WPF, XAML, R, SQL

Nano Precision Products, El Segundo, CA

September 2014 – August 2015 contract

- Project:** Fully automated high speed press system
- Title:** Control Systems Engineer
- Function:** Responsible for development and implementation of automation controls for the high-speed press system, and custom automated systems for use in manufacturing, production, and internal R&D test systems.
- CompactLogix, PanelView Plus, Kinetix 350, EtherNet/IP, Rockwell Studio 5000
 - Cognex – InSight, LabView, Yamaha Robots

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- Voith hydraulic press with Baumuller controller
 - Keyence Laser Marker, 2D barcode reader, Yamaha SCARA Robots
 - Hardware and software integration

Alliance Medical Products, Irvine, CA

July 2013 – May 2014

- Project:** To provide customers with creative solutions to their pharmaceutical or biologics manufacturing requirements
- Title:** Systems Integration Engineer
- Function:** Responsible for automation and systems integration. High speed filling production lines.
- RSLogix 5000, 500, Panel View, Unitronics, OPTO 22, GE S90
 - Cognex - InSight, Checker
 - DeviceNet, Modbus
 - Galil, Yaskawa, LinMot motion controllers
 - Hardware and software integration
 - SCADA – Wonderware – monitoring of production environment
 - Pharmaceutical, highly regulated FDA/GMP environment

Brenner Fiedler & Associates, Inc., Riverside, CA

October 2011 - September 2012

- Project:** A wide variety of PLC and motion control projects
- Title:** Senior Automation & Integration Engineer
- Function:** To support the sale of automation components - engineering services.
- OMRON, UNITRONICS PLCs, STI-OMRON, SYSMAC STUDIO
 - SMAC, QUICKSILVER motion control systems
 - INDUSOFT, WPF – C#, Visual Studio.NET 2010
 - PARALLEX microcontrollers, RFID
 - The project startup and debugging over Internet

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Previous Projects:

- Alcoa, Vernon, CA
- Amada America, Inc., La Mirada, CA
- Anheuser-Busch Companies, Van Nuys, CA
- Belco Packaging Systems, Monrovia, CA
- Calnetix Inc., Cerritos, CA
- Carl Creative Systems, Harbor City, CA
- Ensign-Bickford Optical Technologies, Inc., Van Nuys, CA
- Fadal Machining Centers, Chatsworth, CA
- Gillette Company, Santa Monica, CA
- Northrop, Hawthorne, CA
- Kal Kan, Inc., Vernon, CA
- SymTek Systems, Inc., San Diego, CA
- Technicolor, Inc., North Hollywood, CA
- Triconex Corp. USA, Irvine, CA
- Walt Disney Imagineering, Glendale, CA

REFERENCES AVAILABLE UPON REQUEST.

Zygmunt Marchwian

- Los Angeles, CA, US

Contact Information

- rip-96d-xx4@mail.dice.com
- 3102871265

Summary

Control systems design and **integration** and upgrades: Modular, object oriented design based on a customer specification – UML, P&IDs. Hardware and **software** evaluation, testing and support in selecting the optimal **configuration** based on control goals and any budget. Proficient in **PLC engineering** (**programming**, **configuration**, **integration** and trouble shooting of following platforms: Beckhoff, IDEC, Keyence, Mitsubishi, Modicon, Omron, OPTO 22, Panasonic, Rockwell, Siemens, Unitronics, Wago and any IEC 61131-3 compatible etc. **Software** and hardware **integration** into a fully automated system for manufacturing. OPC, XML, **Modbus**, Device Net and other **communication** technologies. Cognex vision system (InSight, Checker), RFID HMI design and **development**, **SCADA**, Condition Monitoring, Data Acquisition Motion Controls: Servo Drives, Steppers, VFDs, Motors, actuators, etc. Documenting projects with AutoCAD **Electrical** and MS Office Suite (Visio). Commissioning and start up and technical support. **Software** design using advanced Structured **Analysis**: Object, Component Oriented **Programming**, **Programming skills** (VB, C#, ADO, SQL, XML, WPF and Visual Studio.NET etc.). Numerical modeling, **computer** simulations of systems and processes, systems optimization and real time systems **programming** and embedded systems, General networking, protocols, wireless and **troubleshooting**. Good **problem** solving, judgment and decision-making **skills**. Have a gift for finding innovative solutions and for being able to integrate all aspects of bringing a concept to reality. Conduct daily activities ethically and in accordance with a team rules of the culture. Management of internal personnel and outside contractors developing a **project**.

Work History

- System **Integration Engineer** Evolution Construction Services

- **Software** Design **Engineer** Jabil
- Systems **Integration**, Automation **Engineer** Technicolor
- Senior Controls **Engineer** Calnetix
- Senior Control Systems **Engineer** Spectrum A Repligen Brand
- Senior **Software Engineering** Energy Vault
- Senior Automation **Integration Engineer** Brenner Fiedler & Associates
- Control Systems **Engineer** Nano Precision Products
- Systems **Integration Engineer** Alliance Medical Products
- System Integrator Belco Packaging Systems
- Principal Automation & **Integration Engineer** Cynvenio Biosystems

Education

- University of **Science** and Technology in Krakow, Poland

Skills

- aromat **plc**

- automated liquid handling systems
- beckhoff
- c more hmi
- c-more hmi
- canopen
- cimrex hmi
- cognex
- compilers
- computer programming
- cx one
- cx-one
- data acquisition
- dcs
- dvt
- factorytalk view
- iec6131 3
- iec6131-3
- imagineering control language
- lab view
- maple systems hmi
- microsoft office
- microsoft visio
- mitsubishi plc
- modicon plc
- national instruments pxi
- ni daq
- ni-daq
- numerical modeling
- omron plc
- opc keeware
- p&id
- pids
- plc programming
- qci motion ctrl
- relational database
- rfid
- rslogix
- rts
- scada
- servo control
- siemens tia v.13 14
- siemens tia v.13-14
- smac motion ctrl
- sql
- step 7

- **stepper motors**
- **tia portal v11**
- **unitronics plc**
- **vfd**
- **visual c#**
- **wago**
- **win xp embedded**
- **wpf development**
- **yaskawa**
- **software engineering** - 7 years
- **control system** - 6 years
- **integration** - 5 years
- **hmi** - 20 years
- **plc** - 20 years
- **c#** - 15 years
- **engineering** - 4 years
- **microsoft visual studio** - 4 years
- **system integration** - 2 years
- **software** - 4 years
- **wpf** - 3 years
- **motion ctrl** - 20 years
- **networking** - 20 years
- **oop** - 20 years
- **sql server** - 20 years
- **visual studio** - 15 years
- **automation** - 4 years
- **production** - 2 years
- **siemens** - 3 years
- **web portals** - 3 years
- **continuous improvement** - 2 years
- **data processing** - 2 years
- **image processing** - 2 years
- **programmable logic controller** - 2 years
- **xaml** - 2 years

Work Preferences

- Desired Work Settings: Remote or On-Site or Hybrid
- Likely to Switch: False
- Willing to Relocate: False
- Travel Preference: 0%
- Work Authorization:
 - US

- Work Documents:
 - US Citizen
- **Security** Clearance: False
- Desired Hourly Rate: 140+ (USD)
- Third Party: False
- Employment Type:
 - Part-time
 - Contract - Independent

Profile Sources

- linkedin: <https://www.linkedin.com/in/zygmunt-marchwian-17165015>
- Dice: <https://www.dice.com/employer/talent/profile/958cdc5924f0ef10fb2e868509445029>